

Candidate Seat No.: _____

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

Bachelor of Science in Information Technology (B.Sc.IT)

Semester-I University Examination March 2018

CA405 - Foundation of Mathematics

Date: 29/03/2018, Thursday

Time: 10:00 a.m. to 1:00 p.m.

Maximum Marks: 70

Instructions:

1. Attempt the two sections in separate answer books.
2. Figure to the right indicate marks.
3. Make suitable assumptions wherever necessary and state them.

SECTION I

Q-1 Do as directed.

- (A) Define : Empty Set, Relation, Square Matrix and Transpose Matrix [04]
(B) $A \cap \phi =$ _____. [01]
(C) If $A = \{1,2\}$ then find power set $P(A)$. [01]
(D) If $A = \{7,8\}$ and $B = \{2,4,6\}$ then find $A \times B$. [01]
(E) Determine whether $A = \begin{bmatrix} 1 & 2 \\ 2 & 3 \end{bmatrix}$ is symmetric matrix or not. [01]
(F) Find $A+B$ and $A-B$. [02]
 $A = \begin{bmatrix} 0 & 1 & 2 \\ 5 & 7 & 6 \end{bmatrix}$ $B = \begin{bmatrix} 5 & 0 & -2 \\ 7 & -6 & 0 \end{bmatrix}$
(G) Determine whether given relationship is function $f: A \rightarrow B$ or not. [01]
 $A = \{1,3,4\}$, $B = \{7,8,9,10\}$ and relation $R = \{(1,7), (1,8), (3,9), (4,10)\}$

Q-2 Answer following questions.

- (A) List and explain different types of relation with example. [04]
(B) If $A = \begin{bmatrix} 2 & 5 & 7 \\ 1 & 2 & 3 \end{bmatrix}$ and $B = \begin{bmatrix} 3 & 2 & 1 \\ 4 & 7 & 9 \\ 5 & 2 & 4 \end{bmatrix}$ then find AB . [04]
(C) If $f: N \rightarrow N$, $f(n) = 5n+2$, then prove that given $f: N \rightarrow N$ is a function. [04]

OR

Q-2 Answer following questions.

- (A) List out four properties of determinant with an example. [04]
(B) $A = \{x \mid x \leq 5, x \in N\}$, $B = \{x \mid 2 \leq x \leq 8, x \in N\}$ and $C = \{x \mid x \leq 3, x \in N\}$
Find i) $A \cup B$ ii) $A \cap B$ and iii) $B - C$ [04]
(C) Find the inverse of $A = \begin{bmatrix} 1 & 2 & 3 \\ 1 & 3 & 5 \\ 1 & 5 & 12 \end{bmatrix}$ [04]

Q-3 Answer following questions. (Attempt any four)**[12]****(A)** State and prove De-Morgan's law of union operation.**(B)** If $A = \begin{bmatrix} 1 & 2 & 3 \end{bmatrix}$, $B = \begin{bmatrix} 2 & 3 & 0 \\ 0 & 4 & 9 \\ 9 & 1 & 0 \end{bmatrix}$ and $C = \begin{bmatrix} 2 \\ 5 \\ 1 \end{bmatrix}$ then find $(AB)C$.**(C)** The population of town is 6000. Out of these 3400 people read Hindustan Times and 2700 people read The Times of India, there are 700 people who read both the paper. Find number of person who do not read any of these two newspapers.**(D)** If $f(x) = \frac{1}{x+1}$, then find $f(-x) - f(x)$ **(E)** Find determinant value $A = \begin{vmatrix} 3 & 0 & 2 \\ 2 & 0 & -2 \\ 0 & 1 & 1 \end{vmatrix}$ **(F)** If $y = f(x) = \frac{ax+b}{cx-a}$, then prove that $x = f(y)$.**SECTION II****Q-4 Do as directed.****(A)** Evaluate following limit**[03]**

$$i) \lim_{x \rightarrow 2} 5x + 2 \quad ii) \lim_{x \rightarrow 3} \frac{x^3 - 27}{x^2 - 9} \quad iii) \lim_{x \rightarrow \infty} \frac{x^2 + 3x + 5}{2x^2 + 5x + 4}$$

(B) Find derivative of following**[04]**

$$i) y = 2x^2 + 13x + 4 \quad ii) y = 2x + 3x^2 \\ iii) y = x^3 \cdot e^x \quad iv) y = 3\sin x - 4\cos x$$

(C) Find integration of following**[04]**

$$i) \int \sin x \, dx \quad ii) \int x^2 \, dx \\ iii) \int (3 - 2x - x^4) \, dx \quad iv) \int e^{2x+3} \, dx$$

Q-5 Answer following questions.**(A)** Check the continuity of function**[04]**

$$f(x) = \frac{x^2 - 4}{x - 2}, \quad 0 < x < 2$$

$$= x + 2, \quad 2 < x < 5$$

$$= 4, \quad x = 2$$

(B) Find the area bounded by curve $y = x^2$ under the X-axis from point $x=1$ to $x=3$.**[04]****(C)** Find the derivative of $f(x) = x$ using definition.**[04]****OR**

Q-5 Answer following questions.

(A) $f(x) = 4x - 5, x \leq 2$ [04]
 $= x - k, x > 2$

Find value of k, if $\lim_{x \rightarrow 2} f(x)$ is exists.

(B) Find derivative of $y = x^2 + 2x + 4$. For what value of x the derivative became zero. [04]

(C) Find $\int_{-1}^1 (x^2 + 2x - 1) dx$ [04]

Q-6 Answer following questions.(Attempt any four) [12]

(A) Find the derivative of $y = \frac{3}{2-5x}$

(B) Find $\lim_{x \rightarrow 0} \frac{\sqrt{2+3x} - \sqrt{2-5x}}{4x}$

(C) Find the derivative of $y = 2x + 10 \cot^{-1} x$

(D) Find $\int x^2 \cdot e^x dx$

(E) Find $\int 5 \sec 4x \cdot \tan 4x dx$

(F) Evaluate $\lim_{n \rightarrow \infty} \frac{(2n+5)(4n-5)(7n+6)(5n+4)}{(6n+3)(2n+3)(7n-9)(8n+3)}$
